

Deep Learning

Week 4

Question: 1

You are training a neural network on a dataset with 5 million samples using Mini-Batch Gradient Descent. The mini-batch size is 500, and each parameter update takes 100 milliseconds. How many seconds will it take to complete 5 epochs of training?

- A) 5,000
- B) 10,000
- C) 2,500
- D) 50,000

Correct Answer: A

Explanation:

$$\text{Steps per epoch} \rightarrow \frac{5000000}{500} = 10000$$

$$\text{Time per step} = 100 \text{ ms}$$

$$\text{Total time for 1 epoch} = 10000 \times \frac{100}{1000} = 1000 \text{ seconds}$$

$$\text{Total time for 5 epochs} = 1000 \times 5 = 5000 \text{ seconds}$$

Question: 2

You are comparing training times using different gradient descent algorithms on a dataset with 1,000,000 data points. Each parameter update takes 2 milliseconds. How many milliseconds longer will Stochastic Gradient Descent take compared to Vanilla (Batch) Gradient Descent to complete 2 epochs?

- A) 4,000,004 ms
- B) 4,000,000 ms
- C) 3,999,994 ms
- D) 3,999,996 ms

Correct Answer: D

Explanation:

Vanila Gradient Descent 1 step per epoch and for two epochs, the time is $2 \times 2 \text{ ms} = 4 \text{ ms}$

SGD takes 1,000,000 steps per epoch and for two epochs, the time is 4,000,000 ms

Difference is 3,999,996 ms

Information for the questions from 3 to 5

Choosing the Right Batch Size in Resource-Constrained Systems is a very crucial step. You are developing a real-time image recognition system for a mobile health application that must detect skin anomalies from photographs. The device has limited memory and battery, so optimization has to be efficient. Initially, stochastic gradient descent was used, but the results were unstable and slow. You are now considering mini-batch gradient descent and analyzing the impact of different batch sizes and optimizers on both convergence and generalization performance.

Question: 3

What is the most practical benefit of using smaller batch sizes on constrained devices?

Options:

- A) Reduces computation time significantly
- B) Increases training accuracy
- C) Minimizes memory usage and reduces overhead
- D) Allows larger models to be trained

Correct Answer : C

Explanation:

Smaller batch sizes consume less memory and are more suitable for devices with hardware limitations.

Question: 4

You reduce the batch size from 4,000 to 1,000. What happens to the number of weight updates per epoch?

Options:

- A) Doubles
- B) Quadruples
- C) Remains constant
- D) Halves

Correct Answer: B

Explanation:

Updates per epoch = Total samples / batch size. Reducing batch size increases updates proportionally.

Question: 5

Which of the following statements are true about mini-batch gradient descent?

Options:

- A) It offers a compromise between computation and accuracy
- B) It prevents gradient vanishing completely
- C) It allows parallelism in training
- D) It can still be prone to overfitting

Correct Answer : A, C, D

Explanation: Mini-batches support parallel training and offer balance but do not guarantee perfect generalization.

Information for questions _6_ to _11__

You are training a neural network on a noisy dataset of grayscale images.

After 5 epochs,

- The loss is reducing very slowly.
- Accuracy plateaus
- Gradient Magnitudes are very small.

Question: 6

What could be the reason for slow learning in this scenario?

Options:

- A) Large Learning rate
- B) Very small gradients
- C) Very high momentum
- D) Incorrect label noise

Correct Answer : B

Explanation: Small gradients cause small updates to weights, resulting in slow learning.

Question: 7

What optimizer would help improve learning in this situation?

Options:

- A) Vanilla Gradient Descent
- B) Mini-Batch SGD
- C) Momentum based Gradient Descent
- D) Adam with bias correction and adaptive learning rate

Correct Answer : D

Explanation:

Adam adapts the learning rate and includes momentum and bias correction to help in situations with small gradients.

Question: 8

If the above model takes 300 steps per epoch, then after 5 epochs the number of weight updates is _____

Fill in the blank: _____

Correct Answer: 1500

Explanation:

300 updates per epoch, and 5 epochs. $300 \times 5 = 1500$ updates.

Question: 9

Which of the following helps in handling small gradients?

Options:

- A) Reducing learning rate
- B) Using Adagrad
- C) Using adaptive optimizers like Adam
- D) Using large batch sizes

Correct Answer: C

Explanation:

Adaptive optimizers adjust the learning rate dynamically based on gradient history.

Question: 10

What are the advantages of using momentum?

Options:

- A) Faster convergence

- B) Larger steps in the wrong direction
- C) Helps escape shallow minima
- D) Avoids oscillation in steep slopes

Correct Answer: A, C, D

Explanation:

Momentum helps carry forward the previous direction of update, speeding up convergence and reducing oscillations.

Question: 11

Which of the following would not help in this scenario?

Options:

- A) Switch to adaptive gradient descent
- B) Add momentum
- C) Reduce learning rate
- D) Normalize input data

Correct Answer: C

Explanation: Reducing learning rate might slow training even more when gradients are already small.

Information for Questions from __12__ to __13__

In training a deep neural network, the optimization technique used is Momentum-Based Gradient Descent. At each iteration, instead of only considering the current gradient, the optimizer also includes a fraction of the previous update to help accelerate learning and escape shallow local minima.

Question: 12

If the learning rate $\eta = 0.01$, momentum coefficient $\gamma = 0.9$, the current gradient at step t is $\nabla w_t = 0.2$, and the previous update was 0.1, then what is the value of the new update?

Options:

- A) 0.11
- B) 0.092
- C) 0.091
- D) 0.12

Correct Answer: B

Explanation:

$$update_t = 0.9 (0.1) + 0.01(0.2) = 0.092$$

Question: 13

A data scientist is analyzing how momentum affects learning speed over time. She initializes a weight $w_0 = 1.0$, and uses the following parameters in momentum-based gradient descent:

Momentum coefficient $\gamma = 0.8$, Learning rate $\eta = 0.05$ and Initial update $update_0 = 0$

She receives gradients over 3 consecutive iterations:

$$\nabla w_1 = -0.5, \nabla w_2 = -0.2, \text{ and } \nabla w_3 = -0.3$$

What is the value of the update at time $t = 3$?

Options:

- A) 0.0172
- B) -0.0172
- C) 0.0216
- D) -0.009

Explanation:

$$\begin{aligned} update_1 &= 0.8(0) + 0.05(-0.5) = -0.025 \\ update_2 &= 0.8(-0.025) + 0.05(-0.2) = -0.03 \\ update_3 &= 0.8(-0.03) + 0.05(-0.3) = -0.009 \end{aligned}$$

Question: 14

What are the benefits of using mini-batch over full batch.

Options:

- A) Less memory usage
- B) More frequent weight updates
- C) Higher computational cost
- D) Better generalization

Correct Answer: A, B, D

Explanation:

Mini-batch leads to faster convergence and regularization through noise.

Information for the questions from __15__ to __17__

This is regarding oscillations in training a deep neural network. You're training a 6-layer neural network on a regression dataset. The model experiences oscillations in loss after epoch 10, and the model jumps around minima without settling.

Question: 15

What is a likely cause of oscillations?

Options:

- A) Too low learning rate
- B) Batch size too small
- C) Too high learning rate
- D) No dropout

Correct Answer : C

Explanation:

A high learning rate can cause the optimizer to overshoot the minima.

Question: 16

Which technique helps reduce oscillations?

Options:

- A) Momentum
- B) Adagrad
- C) Weight decay
- D) None of the above

Correct Answer: A

Explanation:

Momentum smoothenes updates and reduces oscillations.

Question: 17

Which optimizer allows you to peek ahead before computing the gradient?

Options:

- A) Adam
- B) Vanilla SGD
- C) Nesterov Accelerated Gradient
- D) Adagrad

Correct Answer: C

Explanation: NAG looks ahead using current velocity before computing gradient.

Question: 18

What happens if momentum is set to 1?

Options:

- A) Model stops updating
- B) Model overshoots and diverges
- C) Model converges quickly
- D) No effect

Correct Answer: B

Explanation: High momentum values can cause divergence.

Question: 19

Which of the following are the advantages of mini-batch gradient descent over SGD:

Options:

- A) Reduces variance of updates
- B) Requires fewer epochs
- C) Faster convergence
- D) More computation per update

Correct Answers : A, C

Explanation: Mini-batch smooths updates and balances accuracy and performance.

Information for the questions from _20_ to _21_

You are building a recommendation engine that learns user preferences in real-time. The model must adapt quickly, avoid local minima, and maintain stability during updates. You've observed that a fixed learning rate fails to generalize well. You're experimenting with line search and different decay schedules to improve adaptability of the training algorithm.

Question: 20

What does the line search algorithm aim to optimize at every step of training?

- A) Batch size
- B) The cost function value along the gradient direction
- C) Momentum term
- D) Validation accuracy

Correction Answer: B

Explanation:

Line search dynamically finds the optimal step length along the descent direction.

Question: 21

What is the key computational disadvantage of applying line search in every update?

Options:

- A) May overfit the data
- B) Many more computations in each step.
- C) Doesn't converge
- D) Reduces gradient magnitude

Correct Answer: B

Explanation: Each update involves extra computations to evaluate multiple learning rates.

Question: 22

Which of the following schedules typically require setting two hyperparameters?

Options:

- A) Exponential decay
- B) $1/t$ decay
- C) Constant learning rate
- D) Step decay

Correct Answer: A, B, D

Explanation: These schedules involve setting initial rate and decay control.

Question: 23

Exponential decay adjusts learning rate using which formula?

Options:

A) $\eta = \frac{\eta_0}{(1+kt)}$

B) $\eta = \eta_0^{-kt}$

C) $\eta = \eta_0 \log(t)$

D) $\eta = \eta_0 + t$

Correct Answer: B

Explanation: Exponential decay scales learning rate by a decay factor raised to the power of time.

Question: 24

Learning rate decay is typically used to:

Options:

A) Fine-tune the model toward the end of training

B) Avoid oscillation near minima

C) Eliminate the need for momentum

D) Control the impact of noisy gradients

Correct Answer: A, B, D

Explanation: Smaller rates reduce the chance of overshooting and refine convergence.

Question: 25

In step decay, the learning rate changes at _____ intervals.

Options

A) Predefined

B) One

C) Random

D) None of the above

Answer: A

Explanation: The schedule reduces learning rate at fixed epochs or steps.

Question: 26

If you have 100,000 samples and batch size is 10,000, how many parameter updates happen in one epoch?

Options:

- A) 10
- B) 100
- C) 1000
- D) 1

Correct Answer: A

Explanation: $100000 / 10000 = 10$ updates per epoch.

Question: 27

If $N = 60,000$ and batch size $B = 5,000$, the number of weight updates per epoch = ____.

Fill in the blank _____

Correct Answer: 12

Explanation:

$60000 / 5000 = 12$ updates per epoch.

Information for the questions from __28__ to __29__

You are training a neural network on a large dataset using Nesterov Accelerated Gradient Descent, a variant of momentum-based optimization. Unlike standard momentum, which updates parameters based on the past velocity and current gradient, NAG uses a look-ahead step to compute the gradient — effectively asking “where are we going to be next?” and computing the gradient there.

Question: 28

Suppose you're using Nesterov Accelerated Gradient and are at time step t . The current gradient at the look-ahead position is $\nabla w_{look} = 0.3$, the previous velocity (update) is $update_{t-1} = 0.2$, and the hyperparameters are: $\gamma = 0.8$, $\eta = 0.05$.

What is the value of the current update $update_t$?

Options:

- A) 0.175

- B) 0.195
- C) 0.18
- D) 0.31

Correct Answer: A

Explanation:

$$update_t = 0.8(0.2) + 0.05(0.3) = 0.175$$

Question: 29

You're optimizing a neural network with NAG. At iteration t , you have: Current weight: $w_t = 1.0$, Previous update: $update_{t-1} = 0.25$, $\gamma = 0.9$, $\eta = 0.01$, Gradient at look-ahead position: $\nabla w_{look} = -0.5$.

Using the NAG update rule, what is the value of the updated weight w_{t+1} ?

Options:

- A) 0.78
- B) 0.775
- C) 0.79
- D) 0.77

Correct Answer: A

Explanation:

$$update_t = 0.9(0.25) + 0.01(-0.5) = 0.22$$
$$w_{t+1} = 1.0 - 0.22 = 0.78$$